$\boxed{\text{TP 3 KDE}}$

- obligatory
- individual work.
- penalty -20% for each day of delay

Ex3

$TRN = \{\mathbf{x}_i\}$ $i = 1 \dots n$ $\quad$ x = instance for classification

1) Split TRN by class/label

for h in {...}

2) For each class $c \in C$

$\rightarrow$ prior : $p(c) = \frac{n_c}{n}$ (in TRN) $\left(= \frac{\#ex \in c}{\#ex \in TRN} \right)$

$$\rightarrow \underline{p(\mathbf{x}|c)} = \frac{1}{n_c \underbrace{h^d}_{V}} \sum_{i=1}^{n_c} \prod_{j=1}^{d} k\left(\frac{x_j - x_{ij}}{h}\right)$$

posterior $p(c|\mathbf{x}) \propto p(\mathbf{x}|c) \cdot p(c)$

3) Return c with max posterior

Compute accuracy

$h : \{0.3, 0.4, 0.5)\}$

IRIS data set

$\boxed{\text{Ex 2, Ex 3}}$

training set : iris data

test set : create a regular set of points based on iris data (as in ex1)